Puneeth Kumar Amudala

AI Data Engineer
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Professional Summary

AI Data Engineer with 5 years of experience building cloud pipelines, lakehouse models, and AI retrieval workflows with Python, SQL, and PySpark across AWS, Azure, and GCP. Reduced median catalog publication time from six to two hours and BigQuery bytes scanned by 35%. Experience spans retail catalog, incident reporting, and customer analytics, with Agile and Scrum delivery and stakeholder collaboration.

Professional Experience

Walmart Remote
AI Data Engineer, GCP Catalog Data Platform *May 2026 - Present*
Retail supplier catalog and onboarding platform

- Build Airflow-orchestrated API ingestion pipelines with pagination, retries, failure alerts, and PySpark normalization to load about 50,000 daily supplier product rows into BigQuery, reducing median arrival-to-publication time from six to two hours for catalog analysts.
- Reduce bytes scanned by daily BigQuery reporting queries by 35% through partition filtering, clustering, and incremental transformations.
- Preserve product history and a current catalog view by deduplicating supplier-SKU updates by source version before dbt merges into BigQuery.
- Validate supplier updates with Java services and dbt checks, publish events to Kafka for PySpark processing, and route invalid attributes and categories to catalog analysts for correction.
- Build supplier-onboarding workflows with Python, FastAPI, LangGraph, and MCP tools for taxonomy lookup and duplicate detection; validate attributes with Pydantic and retain source references for analyst review.

Grand Valley State University Allendale, MI
AI DevOps Engineer (Azure Data and RAG Pipelines) *Jan 2026 - May 2026*
Financial-literacy AI assistant for students

- Led application delivery for a 6-person cross-functional Agile team building a financial-literacy assistant for students; mentored undergraduate developers through code reviews.
- Reduced hands-on preparation time for lesson updates threefold by automating ingestion with Azure Data Factory, ADLS Gen2, and Python.
- Avoided rebuilding the full content index by using content hashes to identify changed lessons and update only their embeddings.
- Enabled keyword and semantic retrieval of financial-literacy lessons in Azure AI Search, with query rewriting to support students' follow-up questions.
- Deployed containerized ingestion services on Kubernetes with health checks and restart policies for automatic container recovery, plus resource limits for recurring content updates.

AIG Remote, NC
AI Data Engineer, Databricks on AWS *Jun 2024 - Jun 2025*
Incident data and document workflows for reporting teams

- Built Auto Loader and PySpark pipelines to consolidate about 100,000 daily incident-update rows from four S3 feeds into Bronze Delta Lake tables for reporting teams.
- Cut incident-report job runtime by 10% by tuning PySpark joins; also compacted small Delta files, adjusted job-cluster sizing, and investigated failed runs to identify root causes.
- Built incremental dbt reporting models in Databricks and Snowflake, using source-to-target reconciliation and quarantine gates to keep invalid records out of published tables; scheduled Databricks models with Workflows.
- Made incident evidence retrievable for Amazon Bedrock reports by connecting S3 and Lambda document ingestion to OpenSearch vector indexing.
- Standardized incident updates with Scala-based Spark transformations and merged validated records into Delta Lake tables for downstream reporting.

Cognizant Pune, India
Data Engineer *Nov 2021 - Dec 2023*
Customer data platform for churn and revenue analytics

- Consolidated 2M+ customer records across six sources with Airflow-orchestrated AWS Glue and PySpark pipelines for churn and revenue analytics teams.
- Reduced average reporting-query runtime by 30% through Redshift distribution and sort-key tuning and daily usage aggregates.
- Preserved changes in customer plans and contracts by designing Amazon Redshift fact tables and SCD Type 2 customer dimensions.
- Migrated customer history from Teradata to Amazon Redshift using Informatica mappings and source-to-target reconciliation to validate historical records.
- Kept invalid customer records out of analytics tables with Great Expectations validation and quarantine gates; automated checks in Jenkins CI/CD and routed failed records for correction.

Technical Skills

Languages and Processing: Python, SQL, PySpark, dbt
Data Platforms: Databricks, Delta Lake, BigQuery, Redshift, Snowflake
Pipelines and Cloud: Apache Airflow, AWS Glue, Azure Data Factory, S3, ADLS Gen2, Kafka
Modeling and Quality: Dimensional Modeling, SCD Type 2, Incremental Processing, Great Expectations
AI Data and Agents: RAG, Azure AI Search, OpenSearch, Amazon Bedrock, LangGraph, MCP, FastAPI
Delivery: Docker, Kubernetes, Terraform, Git, CI/CD

Projects

Flight Data Pipeline with Airflow and Snowflake
github.com/Puneeth0106/Realtime-flight-data-engineering

- Built a Python/Airflow pipeline scheduled every 30 minutes to transform OpenSky data through local Bronze, Silver, and Gold layers and load aggregates into Snowflake for analysis of flight activity by country; keyed MERGE operations prevent duplicate rows when a window is rerun.

AWS Cost Optimizer
github.com/Puneeth0106/Aws-cost-optimizer

- Built a Streamlit dashboard combining billing trends with inventory across seven AWS resource categories to help users identify potential cost-saving opportunities; added a LangGraph advisor with six read tools for cost recommendations and defined the serverless backend in Terraform.

Publications

Co-author, Soil Moisture Prediction and Benchmark Reliability: A Review with Multi-Site SMAP–ISMN Analysis. *International Journal on Advances in Intelligent Systems, vol. 19, nos. 1–2, pp. 64–73, 2026.*
`https://www.thinkmind.org/library/IntSys/IntSys_v19_n12_2026/intsys_v19_n12_2026_5.html`

Education

Grand Valley State University Allendale, MI
Master of Science (M.S.), Applied Computer Science | GPA: 3.7 *Jan 2024 - May 2026*
Graduate Badges: Database Management, Data Analytics

Certifications

Databricks Certified Data Engineer Associate	May 2025
Microsoft Certified: Fabric Data Engineer Associate (DP-700)	Oct 2025
Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)	Jan 2026
Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)	Aug 2026
Microsoft Certified: SQL AI Developer Associate (DP-800)	Sep 2026